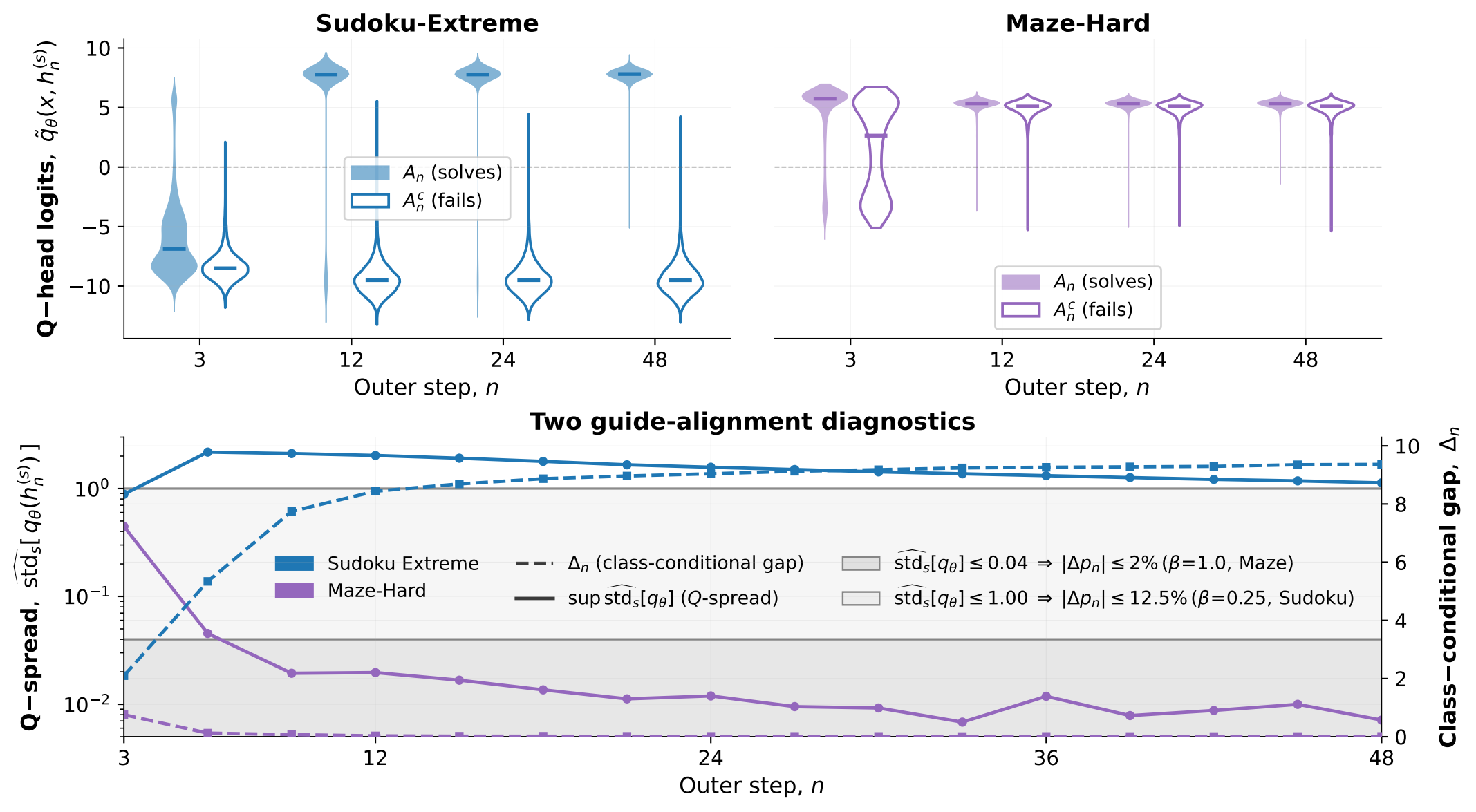

Sudoku-Extreme
Maze-Hard
Q−head logits, $\tilde{q}_\theta(x, h_n^{(s)})$
A_n (solves)
A_n^c (fails)
Outer step, n
Two guide-alignment diagnostics
Q−spread, $\widehat{\mathrm{std}}_s[\, q_\theta(h_n^{(s)})\,]$
Class−conditional gap, Δ_n
Sudoku Extreme
Maze-Hard
Δ_n (class-conditional gap)
$\sup \widehat{\mathrm{std}}_s[q_\theta]$ (Q-spread)
$\widehat{\mathrm{std}}_s[q_\theta] \le 0.04 \Rightarrow |\Delta p_n| \le 2\%$ (β=1.0, Maze)
$\widehat{\mathrm{std}}_s[q_\theta] \le 1.00 \Rightarrow |\Delta p_n| \le 12.5\%$ (β=0.25, Sudoku)